

VLAN ConfigurationLab

1) **Title:** VLAN Configuration using Cisco Packet Tracer

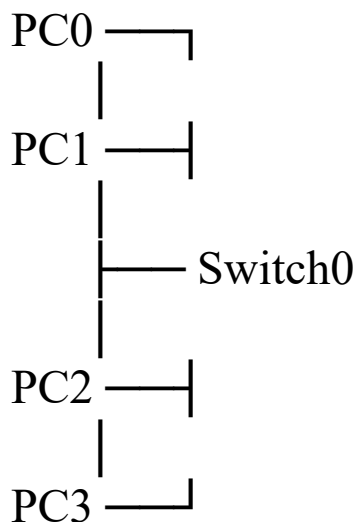
2) **Aim / Objectives :**

To configure VLANs on a switch and establish communication between devices belonging to the same VLAN while isolating devices in different VLANs.

3) **Devices Used :**

- 1 Switch (2960)
- 4 PCs
- Copper Straight-through Cables

4) **Topology :**



5) **VLAN Configuration :**

VLAN ID	VLAN Name	Devices
10	HR	PC0, PC1
20	FINANCE	PC2, PC3

6) IP Addressing Table:

Devices	IP Address	Subnet Mask
PC0	192.168.10.2	255.255.255.0
PC1	192.168.10.3	255.255.255.0
PC2	192.168.20.2	255.255.255.0
PC3	192.168.20.3	255.255.255.0

7) Port Assignment:

Switch Port	Device	VLAN
Fa0/1	PC0	VLAN 10
Fa0/2	PC1	VLAN 10
Fa0/3	PC2	VLAN 20
Fa0/4	PC3	VLAN 20

8) Procedure / Steps :

Step 1: Add Devices

- Add one 2960 Switch.
- Add four PCs.
- Arrange the devices on the workspace.

Step 2: Connect Devices

Use Copper Straight-through cables.

Connect:

- PC0 → Switch Fa0/1
- PC1 → Switch Fa0/2
- PC2 → Switch Fa0/3
- PC3 → Switch Fa0/4

Step 3: Configure IP Addresses

PC0

IP Address: 192.168.10.2

Subnet Mask: 255.255.255.0

PC1

IP Address: 192.168.10.3

Subnet Mask: 255.255.255.0

PC2

IP Address: 192.168.20.2

Subnet Mask: 255.255.255.0

PC3

IP Address: 192.168.20.3

Subnet Mask: 255.255.255.0

Step 4: Create VLANs

Open Switch CLI and enter:

enable

configure terminal

vlan 10

name HR

exit

vlan 20

name FINANCE

exit

Step 5: Assign Ports to VLAN 10

interface range fa0/1 - 2

switchport mode access

switchport access vlan 10

exit

Step 6: Assign Ports to VLAN 20

interface range fa0/3 - 4

switchport mode access

```
switchport access vlan 20
exit
```

Step 7: Verify VLAN Configuration

```
show vlan brief
```

Expected Output:

```
VLAN 10 HR      Fa0/1 Fa0/2
VLAN 20 FINANCE Fa0/3 Fa0/4
```

Step 8: Test Connectivity

Same VLAN Communication

From PC0:

```
ping 192.168.10.3
```

Expected Result:

```
Successful Reply
```

From PC2:

```
ping 192.168.20.3
```

Expected Result:

```
Successful Reply
```

Different VLAN Communication

From PC0:

```
ping 192.168.20.2
```

Expected Result:

```
Request Timed Out
```

Because devices in different VLANs cannot communicate without a router or Layer 3 switch.

9)Result :

VLANs were successfully configured on the switch. Devices within the same VLAN communicated successfully, while communication between different VLANs was restricted.

10) Concepts Learned :

- VLAN Creation
- Network Segmentation
- Switch Port Assignment
- Broadcast Domain Separation
- VLAN Isolation
- Basic Network Security

11) Cybersecurity Relevance :

- VLANs improve network security by isolating departments and users.
- They reduce the attack surface within a network.
- They help prevent unauthorized access between different network segments.
- VLAN segmentation is commonly used in enterprise environments to separate HR, Finance, IT, and Guest networks.

12) Screenshot :

